



Sequential Estimation of Remaining Useful Life Through VAE-Based Deep Reinforcement Learning With Reward Shaping

Hima Soni¹ · Vibha Patel² · Arnav Modanwal³ · Hepil Italiya⁴ · Aastha Patel⁵

Received: 8 January 2026 / Revised: 24 July 2026 / Accepted: 24 August 2026

© The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2026

Abstract

In industrial operations, reducing unscheduled machinery downtime, extending equipment life, and ensuring economical operations are crucial for reliability. Furthermore, to improve sequential Remaining Useful Life (RUL) estimation through latent-space policy optimization, the current study introduces a novel framework that integrates Variational Autoencoders (VAE) as surrogate environments with actor-critic Deep Reinforcement Learning (DRL). The VAE enhances representation quality by reducing sensor data to compact latent features that maintain the underlying degradation dynamics. Two actor-critic DRL algorithms, Twin Delayed Deep Deterministic Policy Gradient (TD3) and Proximal Policy Optimization (PPO), are employed to learn sequential latent-space optimization policies that iteratively refine degradation trajectories within the learned surrogate environment. The comparative evaluation shows that optimization strategy meaningfully shapes learning stability and downstream prediction accuracy across reward formulations, with the on-policy approach yielding more consistent convergence in 11 of 12 configurations. However, the Weighted Cosine Similarity reward produced comparable RUL prediction accuracy, with large Cohen's d effect sizes ($|d| > 0.8$) in 10 configurations. The best-performing variant achieved an RMSE of 0.1262, thereby outperforming conventional supervised sequence-learning baselines, including LSTM and MLP, by 26.7–40.3%. The proposed architecture is computationally scalable, data efficient, and adaptable to a variety of degradation patterns. These results demonstrate the potential of combining sequential latent-space policy optimization with generative representation learning to provide a modular and efficient sequential RUL estimation method for challenging industrial scenarios.

Keywords Predictive maintenance · Remaining useful life (RUL) · Deep reinforcement learning · Reward design · Reliability engineering

Editor: Panagiotis Papapetrou.

Extended author information available on the last page of the article

1 Introduction

Maintenance is crucial for the operability, reliability, and economic viability of industrial systems. The conventional reactive and preventive maintenance approaches have gradually been replaced by predictive maintenance (PdM) methods. PdM employ sensor-driven diagnostics to anticipate equipment malfunctions and optimize maintenance timelines over the past decade. The benefits of PdM include reduced unplanned downtime, lower operational costs, and extended asset lifespans (Crespo et al., 2020). However, these benefits do not always translate into solutions that balance proactive interventions with real-time adaptability in dynamic, complex industrial environments.

The capabilities of PdM systems have been significantly improved by recent advances in data-driven techniques, particularly in Machine Learning (ML) and Deep Learning (DL). Convolutional Neural Networks (CNNs) and LSTM models are widely used for time-series data analysis and degradation signature learning. These models can be useful, but they are usually less robust when working with sparsely annotated data, under different operating settings, and when generalizing to previously encountered scenarios. Using a limited set of activation functions and an adaptive learning algorithm, lightweight systems like Extreme Learning Machines (ELM) have also been investigated for their rapid fault-classification capabilities in sensitive infrastructure (Dhini et al., 2022). Reinforcement Learning (RL) and generative models have recently emerged as promising approaches for improving adaptive sequential optimization in data-driven prognostics. Variational Autoencoders (VAEs) are particularly well suited for learning compact latent representations of system dynamics that preserve the underlying degradation behavior while reducing the complexity of high-dimensional sensor observations (Kingma and Welling, 2013). When combined with learned surrogate environments, RL agents iteratively optimise latent state transitions, allowing sequential refinement of degradation trajectories instead of independent point-wise predictions. This formulation enables adaptive optimization over the evolution of degradation trajectories while naturally incorporating different reward formulations within a common learning framework. Prior research also demonstrated the usefulness of RL in cost-sensitive industrial situations because of its ability to adapt to evolving system behavior (Rasay et al., 2025; Xanthopoulos et al., 2019).

The proposed framework formulates RUL estimation as a sequential optimization problem within a learnt latent state space, in contrast to traditional supervised sequence models that directly minimize prediction error through a preset regression objective. The policy selects a latent action, monitors the current degradation state, and obtains reward-driven feedback based on the prediction trajectory that results at each timestep. As a result, optimization is carried out not over independent point-wise predictions but over the evolution of the complete latent trajectory. Additionally, this formulation preserves temporal consistency throughout the prediction process and allows for the systematic study of various reward mechanisms under a common learnt representation.

Substantial gaps in PdM research still remain despite recent advancements. Specifically, there is still a lack of research on using latent representations obtained from VAE as differentiable surrogate environments for sequential policy optimization. In addition, the majority of RL-based prognostic methods now in use rely on generic or manually designed reward formulas, which might result in inconsistent learning behavior and unstable performance under different operating conditions. There are currently few systematic research examin-

ing how algorithm selection and incentive design affect sequential Remaining Useful Life (RUL) estimation within a learned latent dynamics (Jiménez et al., 2023).

This research introduces a modular framework for sequential Remaining Useful Life (RUL) estimation that combines a Variational Autoencoder (VAE)-based surrogate environment and actor-critic Deep Reinforcement Learning (DRL). The key methodological contribution is to reformulate RUL estimation as a sequential latent-space policy optimization problem, in which the policy learns a sequence of latent control actions to progressively refine degradation trajectories by maximizing cumulative reward rather than directly minimizing point-wise prediction error. Unlike traditional reinforcement learning approaches for maintenance scheduling, the proposed framework is solely focused on latent-space policy optimization for sequential RUL estimation. To investigate the influence of different policy optimization approaches, two actor-critic algorithms, PPO and TD3, are evaluated inside the same learnt latent environment. The framework also looks into the impact of twelve reward-shaping mechanisms on learning stability, convergence behaviour, and trajectory prediction performance. The combination of generative representation learning with reinforcement learning yields a scalable, data-efficient, and interpretable solution to sequential RUL estimation in industrial environments. Furthermore, the proposed methodology demonstrates how latent-space policy optimization, together with a learnt surrogate environment, can improve sequential RUL estimation under non-stationary degradation dynamics.

2 Literature Review

PdM is now more vital than ever for improving operational efficiency, extending equipment life, and minimizing maintenance costs. This overview describes the evolution of PdM techniques from traditional rule-based systems to complex data-driven, deep learning (DL), and RL frameworks, emphasizing the ongoing problems that drive the current endeavor.

2.1 Background on PdM and Remaining Useful Life Estimation

Early PdM techniques were mainly reliant on statistical process management and rule-based diagnostics. These strategies performed well in organized situations, but they were insufficient to deal with unanticipated failure types and system unpredictability. More adaptable modelling became possible with the advent of data-driven Machine Learning (ML) techniques, especially for component degradation prediction and multivariate sensor stream analysis. A benchmark for simulating damage propagation in aero-engines, the C-MAPSS dataset (Saxena et al., 2008), allowed for the validation of Remaining Useful Life (RUL) prediction models under various degradation trajectories.

In order to advance RUL estimation, DL architectures have become essential. While hybrid models that combined CNNs and LSTMs (Al-Dulaimi et al., 2019) more successfully captured spatial-temporal dependencies, CNNs applied to the C-MAPSS dataset (Li et al., 2018) demonstrated better accuracy. However, because there is little validation across many operating regimes, these techniques often have problems with generalization. Although domain-specific calibration remained crucial, transfer learning (Fan et al., 2020) attempted to close this gap by modifying pre-deployment models to post-deployment data. While stochastic RNNs in conjunction with VAEs (Tanzeem and Moghaddass, 2025) and

variational autoencoder (VAE)-based frameworks (Su et al., 2020) provided better temporal modelling, they did not integrate with the latent-space policy optimization processes necessary for adaptive policy learning. Specialised deep networks and ensemble designs have further improved prediction performance. Domain-specific efficacy was shown by techniques like EDL-MO (Ma et al., 2021), regularised E-LSTM for rolling bearings (Liu et al., 2021), and fault detection CNNs for rotating machinery (Caponetto et al., 2019). For steam turbines in particular, lightweight alternatives such as ELM-RBF (Dhini et al., 2022) provided quick inference with competitive fault diagnostic accuracy. Scalability is still limited by dependency on large labelled datasets and significant retraining under dynamic situations, notwithstanding these advances.

Prognostics have been further enhanced by advanced feature engineering approaches such as multiscale bidirectional LSTMs (Gu et al., 2025), stepwise feature selection (SFS), and signal reconstruction by EEMD. However, the deployment of these methodologies across heterogeneous industrial systems is hampered by the significant domain expertise and preprocessing requirements. For capturing relational and spatiotemporal patterns, graph-based methods have drawn interest. Although they lack real-time adaptability and require manual graph creation, models like spatiotemporal knowledge graphs (STKGs) combined with graph attention pooling (Men et al., 2025) successfully simulate sensor interactions. For RUL prediction in induction motors, nonlinear degradation indices employing similarity-based ensembles have also demonstrated promise (Kang et al., 2024). However, addressing diverse data quality and standardizing across sensor types are still unresolved issues.

Long-term cost optimization and adaptability are still issues with traditional supervised learning techniques. Although they provide predictive power, probabilistic CNN frameworks (Lee and Mitici, 2023), DL ensembles (Wang et al., 2024), and multicellular LSTM networks (Xiang et al., 2021) have limited generalization and large computational overheads. These limitations emphasize the need for learning paradigms that facilitate adaptive latent-space policy optimization with less labelled input and preprocessing and function in dynamic environments.

2.2 Advanced Modeling Approaches and Maintenance Optimization

More flexible modelling is required for modern industrial systems, which are often characterised by intricate deterioration stages and interrelated components and require more flexible modelling. by intricate deterioration stages and interrelated components. Although they rely on strong prior assumptions, probabilistic degradation processes, such as the two-stage inverse Gaussian model (Yan et al., 2025), effectively managed degradation transitions. Wiener processes with threshold error have also been explored to include maintenance threshold variability (Ren et al., 2024), but real-time adaptation and integration into learning-based pipelines were challenges for these approaches. Acceleration factor approaches appropriate for sporadic inspections have been shown to improve field-based RUL estimates (Jeong et al., 2025), but their high computing requirements make them impractical for scenarios with limited data. Semi-regenerative models are used in condition-based maintenance for multi-component systems (Zhang et al., 2025), which takes stochastic dependencies into account, but it does not seamlessly integrate with deep learning. Although it is distinct from deep policy-learning frameworks, inventory-aware joint optimization (Zhang et al., 2018) shows cost-effectiveness under resource restrictions.

Strategies for dynamic maintenance are changing. Although they provide for coarse-grained flexibility, hybrid systems that combine long-term LSTM forecasts with real-time rule-based changes (Yaqiong et al., 2025) are finding it difficult to adjust to rapidly changing priorities. RUL-based replacement timing (Choo et al., 2025) have been used to handle multistage production scenarios with inter-machine dependencies, however many of these models were static and unresponsive to operational changes. Interpretability and diagnostics in real-world settings were improved by recent developments such SHAP-based explainable models with IoT integration (Cardoso et al., 2025). Emerging concepts like Cognitive Digital Twins (CDTs) (Iyer et al., 2025) and e-maintenance frameworks (Ucar and Qiu, 2005) offer scalable integration of diagnostics, AI and process-level monitoring. While e-maintenance systems connect wireless data collection with condition monitoring, CDTs provide bottleneck detection and explainable anomaly analysis. Long-term PdM success is still largely dependent on human factors, as evidenced by fuzzy SERVQUAL assessments of Total Productive Maintenance (TPM) (Mouhib et al., 2025).

2.3 Integration of Generative Models and RL for Prognostics

Although they increase confidence estimates, uncertainty quantification methods employing Bayesian neural networks (BNNs) (Benker et al., 2021) and Monte Carlo dropout-based CNNs (Mitici et al., 2023) confront scalability and computational cost issues. A paradigm change from passive prediction to active policy optimization is provided by RL. Under demand variability, RL-based production control has outperformed classic techniques (Xanthopoulos et al., 2019), and sustainable upgrade planning has been influenced by dynamic programming techniques (Chung et al., 2017). Even with these developments, there is still much to learn about integrating generative modeling with actor-critic RL agents for PdM. Rarely do current studies use latent representations obtained from VAE as surrogate environments for policy learning, and they do not systematically examine how different incentive designs affect learning dynamics and generalization performance in RUL tasks.

The present study proposes a novel and modular PdM paradigm that combines VAEs with actor-critic DRL to bridge these gaps. In particular, optimal sequential estimation policies are obtained by training PPO and TD3 algorithms inside a learned latent degradation space. The methodical assessment of twelve reward shaping techniques, which offers empirical insights into how reward design affects convergence behavior and policy performance, is a noteworthy contribution. Through this integration, the framework offers a scalable, data-efficient and adaptable solution for intelligent maintenance scheduling in complex industrial environments.

3 Preliminaries

This section describes the theoretical foundations of the suggested framework. The generative modeling of degradation dynamics using VAEs is described first, and then DRL methodologies used for optimal maintenance latent-space policy optimization are discussed.

3.1 Generative Modeling of System Dynamics

High-dimensional sensor data in industrial systems commonly exhibits nonlinear and non-stationary patterns that traditional parametric state-space models challenge to represent. To solve this, a compact latent representation of system dynamics is trained using a Variational Autoencoder (VAE) (Kingma and Welling, 2013). An encoder network that approximates the posterior distribution makes up the VAE. $q_\phi(z|y)$ over latent variables z given inputs y and a decoder network that models $p_\theta(y|z)$ to reconstruct the observations.

The VAE's objective is to optimize the Evidence Lower Bound (ELBO), which achieves a balance between latent space regularization and reconstruction accuracy (Kingma and Welling, 2013):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(z|y)}[\log p_\theta(y|z)] - D_{\text{KL}}(q_\phi(z|y) \parallel p(z)), \quad (1)$$

where D_{KL} denotes the Kullback-Leibler divergence and $p(z)$ is typically a standard Gaussian prior. To allow end-to-end training, the latent variable z is sampled using the reparameterization trick: $z = \mu + \sigma \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

A nonlinear autoregressive framework is used for extending the VAE in order to explicitly incorporate temporal dependencies that are essential for degradation modeling. Specifically, a Nonlinear Autoregressive model with Exogenous Inputs (NARX) (Kelley and Hagan, 2024) is employed in the decoder. The latent codes z are treated as exogenous inputs, allowing the system's future states to be inferred from historical observations:

$$y(t) = F[y(t-1), \dots, y(t-n); z(t-d), \dots, z(t-d-m)], \quad (2)$$

where d is the time delay and F is a nonlinear mapping acquired using neural networks. Sequential degradation processes can be expressively modeled using this hybrid VAE-NARX structure without the need for explicit physical modeling or hand-crafted features.

3.2 DRL as Sequential Optimization Framework for RUL Estimation

After learning the degradation representation, the RUL estimation problem is formulated as a Markov Decision Process (MDP), where the goal is to find an optimal policy π^* that maximizes the expected cumulative reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \quad (3)$$

with $\gamma \in [0, 1]$ as the discount factor and r_t the reward at timestep t . To solve this problem, actor-critic methods are employed due to their efficiency in continuous action spaces.

The first algorithm considered is **PPO** (Schulman et al., 2017), an on-policy method designed to stabilize training by limiting the size of policy updates. The clipped surrogate objective is given by:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad (4)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio and $\hat{A}_t$ is the estimated advantage function.

By penalizing large deviations from the previous policy, PPO encourages stable yet progressive improvement.

The second algorithm, **TD3** (Fujimoto et al., 2018), is an off-policy actor-critic approach particularly suited for high-dimensional and continuous control tasks. TD3 addresses the overestimation bias in value functions by maintaining two independent critic networks and using their minimum value for target computation:

$$y = r + \gamma \min_{i=1,2} Q'_i(s', \mu'(s') + \epsilon), \quad (5)$$

where ϵ is a small random noise added for policy smoothing. Additionally, TD3 delays actor updates and target value updates to improve stability, making it well-suited for complex maintenance scenarios with sparse rewards and noisy sensor signals.

Together, these components VAE-based latent modeling and DRL-based policy optimization form the foundation of a scalable, end-to-end framework for PdM under real-world industrial conditions.

4 Methodology

This section introduces a methodological framework for estimating the remaining useful life (RUL) that is both reliable and adaptable. The proposed architecture combines Deep Reinforcement Learning (DRL) and a deep generative surrogate environment to develop a modular framework for sequential Remaining Useful Life (RUL) estimation. The approach is divided into two main steps: (i) policy learning using DRL for RUL estimation and (ii) surrogate environment modeling based on a Variational Autoencoder (VAE) and a Nonlinear Autoregressive (NARX) structure. Here, the ideas discussed in Section 3 are used in a data-driven, real-world scenario.

4.1 VAE-NARX Surrogate Modeling of Degradation Dynamics

In the initial phase, a VAE is trained using historical multivariate sensor data and related RUL values to create a data-driven surrogate environment. Long-term equipment health evolution can be simulated due to this model, which encapsulates degradation behavior into a structured latent space.

The input vector at each time step is composed of 21 sensor measurements and the ground-truth RUL value, S_t :

$$S_t = [\text{Sensor}_1, \dots, \text{Sensor}_{21}, \text{RUL}_t]. \quad (6)$$

The encoder transforms S_t into a latent representation z_t , modeled as a multivariate Gaussian distribution with mean μ and standard deviation σ . Sampling is performed using the reparameterization trick:

$$z_t = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (7)$$

Following that, the decoder simulates future system states by reconstructing the sensor readings and RUL values from z_t . A NARX-based structure, which forecasts the system's future state based on historical outputs and the latent input, is used to extend the decoder in order to integrate sequential dynamics. This design allows the model to capture both temporal correlations and non-linear degradation trends.

The VAE is optimized by maximizing the standard Evidence Lower Bound (ELBO), defined as:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\phi}(z|x) \parallel p(z)), \quad (8)$$

where D_{KL} denotes the Kullback-Leibler divergence and $p(z)$ is a standard Gaussian prior. Once trained, the VAE decoder operates as a fully differentiable simulation environment, enabling DRL-based interaction without requiring expensive physical experiments.

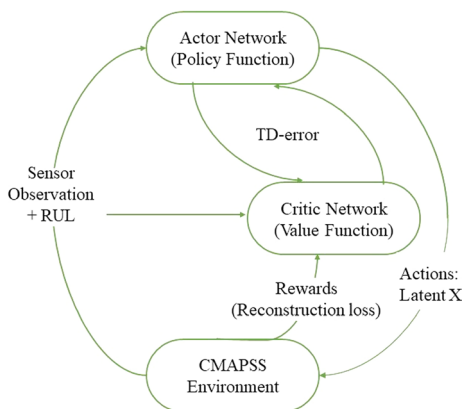
4.2 DRL Framework for RUL Estimation

A DRL agent is trained in the VAE decoder's surrogate environment during the second phase. In order to estimate sequential RUL trajectories, the agent learns to interact with the learned latent space. Rather than directly minimizing point-wise prediction error, the policy progressively refines latent degradation trajectories across episodes through reward-driven sequential optimization. Figure 1 depicts the interaction scheme. The decoder mimics the system behavior resulting from the DRL agent's latent action selection based on learned state representations. This closed interaction loop enables policy learning without requiring explicit degradation models or manual feature engineering.

The components of the DRL framework are defined as follows:

- **State space ($\mathcal{S}$):** The latent vector z_t obtained by the VAE encoder makes up the state s_t at each time step t . The operating profile and degradation history are encoded within this latent representation. A 25-dimensional input space is produced by the complete observation vector, which consists of 21 raw sensor values, three operational settings, and the RUL label.

Fig. 1 Overview of the proposed VAE-DRL framework for sequential latent-space policy optimization in RUL estimation



- Action space ($\mathcal{A}$): The action variable a_t does not represent a physical maintenance intervention. Instead, it serves as a learned latent control variable selected by the reinforcement learning policy to influence the evolution of degradation trajectories within the VAE decoder's latent space. The objective of the proposed framework is therefore sequential latent-space policy optimization for RUL estimation rather than maintenance scheduling. Mapping these latent control actions to physically interpretable maintenance decisions is beyond the scope of the present work and remains an important direction for future research. During policy learning, the agent determines a sequence of latent control actions that guides the degradation trajectory over time, while the decoder converts the resulting latent states into sequential RUL estimates.
- Reward function (r_t): The reward function r_t encourages the agent to learn latent trajectories that reduce sequential RUL estimation error over time. After each latent control action, the surrogate environment predicts the next latent state and its corresponding RUL estimate. The reward is then computed from the absolute difference between the predicted and true RUL values, providing the scalar feedback used by the policy optimization algorithm to maximize expected cumulative reward over the prediction horizon:

$$r_t = -|\text{RUL}_{\text{pred}} - \text{RUL}_{\text{true}}|. \quad (9)$$

To examine the influence of reward design on policy learning, twelve reward formulations were evaluated, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Huber loss, and several hybrid formulations. This comparison provides insight into how different

Table 1 Reward functions and their corresponding mathematical formulations

Sr. No.	Reward function	Equation
1	Huber loss	$\begin{cases} \frac{1}{2}(y - \hat{y})^2, & y - \hat{y} \leq \delta \\ \delta(y - \hat{y} - \frac{1}{2}\delta), & \text{otherwise} \end{cases}$
2	Logcosh loss	$\log[\cosh(y - \hat{y})]$
3	Cosine similarity loss	$\frac{\sum_{i=1}^n y_i \hat{y}_i}{\sqrt{\sum_{i=1}^n y_i^2} \sqrt{\sum_{i=1}^n \hat{y}_i^2}}$
4	MSE loss	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
5	MAPE loss	$\frac{100}{n} \sum_{i=1}^n \left \frac{y_i - \hat{y}_i}{y_i} \right $
6	MAE loss	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
7	MSE + MAE combination	$\alpha \cdot \text{MSE} + (1 - \alpha) \cdot \text{MAE}$
8	Huber + MAE combination	$\alpha \cdot \text{Huber} + (1 - \alpha) \cdot \text{MAE}$
9	Huber + MSE combination	$\alpha \cdot \text{Huber} + (1 - \alpha) \cdot \text{MSE}$
10	MAE + Logcosh combination	$\alpha \cdot \text{MAE} + (1 - \alpha) \cdot \text{Logcosh}$
11	MSE + Logcosh combination	$\alpha \cdot \text{MSE} + (1 - \alpha) \cdot \text{Logcosh}$
12	Weighted cosine similarity	$w \cdot \text{CosineSimilarity}(y, \hat{y})$

reward functions affect policy convergence, learning stability, and generalization performance. The complete list of reward formulations is presented in Table 1.

By integrating learned latent features with reward-driven learning, the proposed VAE-DRL system offers a scalable and adaptable solution for PdM across varying operational conditions. The framework's modularity allows it to generalize across equipment types and sensor modalities, minimizing the reliance on task-specific assumptions or domain heuristics.

4.3 DRL Algorithms for Policy Optimization

To optimize sequential RUL estimation within the learnt surrogate environment, two state-of-the-art actor-critic DRL algorithms, PPO and TD3, were used. Both algorithms learn policies that iteratively optimize latent degradation trajectories via reward-driven interaction with the VAE-based surrogate environment.

4.3.1 PPO

PPO employs a clipped surrogate objective to achieve consistent and conservative policy adjustments. The actor network learns a stochastic policy that converts latent states to continuous latent actions, while the critic estimates the expected return for each latent state. In this approach, both the actor and the critic are represented by two completely connected hidden layers with 64 neurons each, activated by the Tanh function. This design provides a balance between expressiveness and training stability, making it ideal for sequential RUL estimation in learnt non-stationary degradation environments. The PPO agent's conceptual flow is illustrated in Figure 2, while the detailed training steps for integrating PPO with the custom VAE-based environment are presented in Algorithm 1.

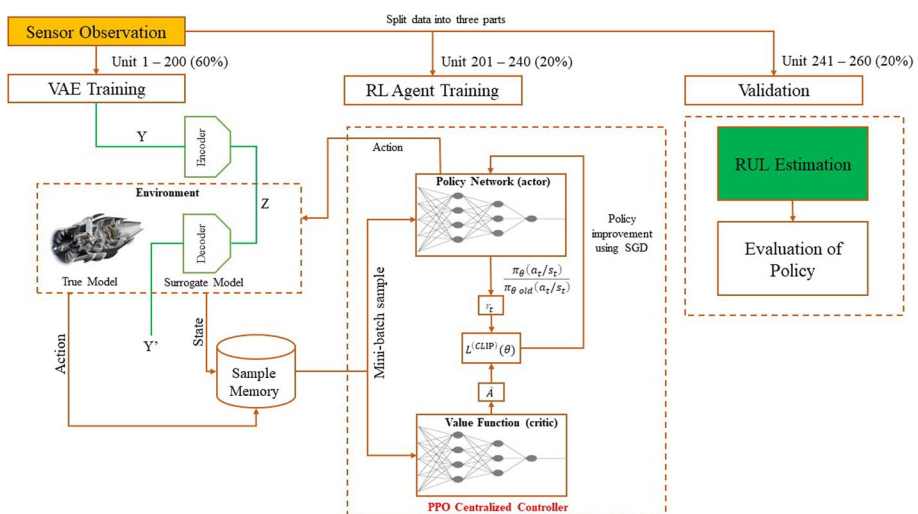


Fig. 2 PPO-based sequential latent-space policy optimization for RUL estimation

```

1: Input:  $df$ , VAE decoder model decoder, engine lives engine.lives, environment
   configuration env_config
2: Policy parameters initialization  $\theta_0$ , value function parameters  $\phi_0$ 
3: Custom environment initialization  $env = \text{CMAPSSEnv}(df, 0, \text{decoder})$ 
4: Wrap the environment in a vectorized environment  $env = \text{DummyVecEnv}([env])$ 
5: for iteration  $k = 0, 1, 2, \dots$  do
6:   Trajectories collection  $\mathcal{D}_k = \{\tau_i\}$  by running policy  $\pi_k(\theta_k)$  in  $env$ 
7:   Rewards-to-go computation  $\hat{R}_t$  using one of the 12 reward functions as per
   Table 1
8:   Computation of advantage estimates  $\hat{A}_t$ 
9:   Computation of policy ratios  $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}$ 
10:  for epoch  $i = 0, 1, 2, \dots$  (until convergence) do
11:    Surrogate loss optimization
12:     $\theta_{k+1} \leftarrow \arg \max_{\theta} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right)$ 
13:  end for
14:  Fit the value function using regression on mean-squared error:
15:   $\phi_{k+1} = \arg \min_{\phi} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \left( V_{\phi}(s_t) - \hat{R}_t \right)^2$  Schulman et al. (2017)
16: end for
17: Output: Optimized latent-space policy  $\pi_{\theta}$ 

```

Algorithm 1 PPO Schulman et al. (2017) with VAE and custom environment

```

1: Input: Data  $df$ , VAE decoder decoder, engine lives engine.lives, environment
   config env_config
2: Initialize critic networks  $Q_1, Q_2$  and actor  $\mu$  with random weights; set target
   networks  $Q'_1, Q'_2, \mu'$  to equal main networks
3: Initialize custom environment  $env = \text{CMAPSSEnv}(df, 0, \text{decoder})$  and replay
   buffer  $R$ 
4: for episode = 1 to  $M$  do
5:   Receive initial state  $s_1$ 
6:   for  $t = 1$  to  $T$  do
7:     Select action  $a_t = \mu(s_t) + \mathcal{N}(0, \sigma)$  and observe  $(r_t, s_{t+1})$  using one of 12
     reward functions from Table 1
8:     Storing  $(s_t, a_t, r_t, s_{t+1})$  in  $R$  and sample minibatch of  $N$  transitions
9:     Compute target action  $\tilde{a}_{t+1} = \mu'(s_{t+1}) + \epsilon$ , with clipped noise  $\epsilon$ 
10:    Update critics by minimizing  $L = (Q_1 - y)^2 + (Q_2 - y)^2$ , where  $y = r +$ 
     $\gamma \min(Q'_1, Q'_2)$ 
11:    if  $t \bmod d = 0$  then
12:      Update actor using deterministic policy gradient and softly update
      target networks
13:    end if
14:  end for
15: end for
16: Output: Optimized latent-space policy  $\mu_{\theta}$ 

```

Algorithm 2 TD3 Fujimoto et al. (2018) with VAE and custom environment

4.3.2 TD3

TD3 extends the deterministic DDPG framework by addressing common overestimation issues in Q-value prediction. It incorporates three key innovations: twin Q-networks to compute the minimum of predicted values, delayed updates to the actor network and target policy smoothing. The actor network consist three hidden layers and each critic network consists of two hidden layers; exact architectural parameters are elaborated in Section 4.6. TD3's architecture is tailored to produce more stable policy updates in high-dimensional continuous latent action spaces. Figure 3 shows the TD3-based flow and Algorithm 2 outlines the full training process within the custom environment.

Although PPO and TD3 employ different optimization strategies, both operate within the same learned latent surrogate environment, allowing a controlled comparison of policy optimization mechanisms under identical state representations and reward formulations.

4.4 Model Training and Evaluation

The DRL agent is trained through repeated interaction with the learned surrogate environment across a series of episodes, progressively optimizing its latent-space policy using reward-driven feedback. During each episode, the agent observes the current latent state, selects a latent action using its learned policy, receives a reward based on the resulting RUL estimation error, and transitions to the next latent state through the VAE decoder. This interaction loop enables the policy to progressively refine latent degradation trajectories, resulting in temporally consistent sequential RUL estimates without relying on explicit degradation models. The reward function that drives this learning is chosen out of a set of twelve options (described in Table 1), which include loss-based functions including MSE, MAE, Huber loss and hybrid formulations. These reward formulations enable systematic investigation of how different optimization objectives influence convergence behavior, trajectory prediction accuracy, and policy stability. Successive iterations also help the agent to

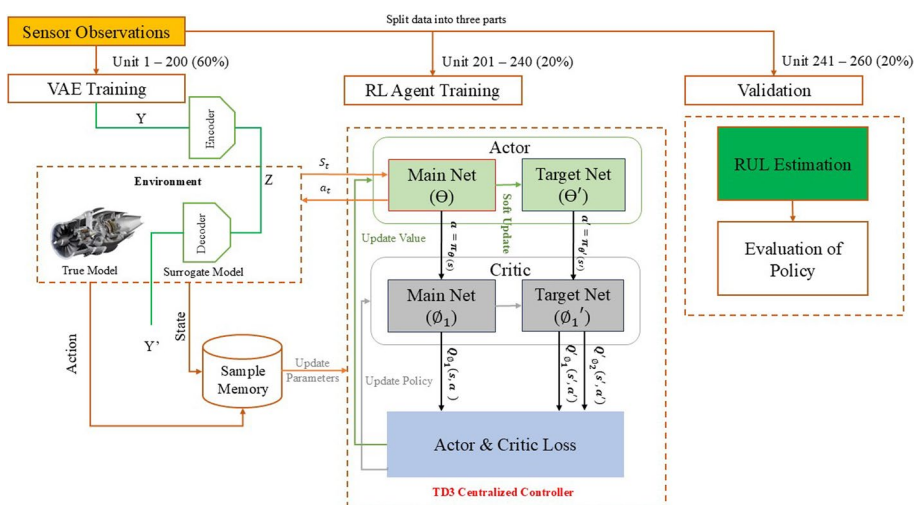


Fig. 3 TD3-based sequential latent-space policy optimization for RUL estimation

minimize the RUL prediction error and thereby generating robust sequential RUL trajectories across different degradation stages.

Evaluation is conducted using standard regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2), computed over the predicted RUL trajectories across three evaluation seeds (42, 124, 777) to assess result variability. To quantify the practical magnitude of performance differences between PPO and TD3, Cohen's d effect sizes were computed on RMSE values, where $|d| > 0.8$ indicated a large, practically meaningful effect. The NASA asymmetric scoring function (S-score) is not reported, as it is designed for single-point RUL prediction at a fixed evaluation horizon, whereas the proposed framework produces sequential trajectory-level estimates that are not directly compatible with the single-point evaluation protocol. Additionally, to establish the value of the latent-space policy optimization framework, comparisons are conducted against supervised learning baselines including Linear Regression, Multi-Layer Perceptrons, and LSTM networks trained on the same features and evaluation split. The average and standard deviation of cumulative episode rewards are also monitored to evaluate the stability and consistency of the learned latent-space policies. Using a common latent representation and identical training conditions for both PPO and TD3 ensures that observed performance differences primarily reflect the policy optimization strategy and reward formulation rather than differences in feature representation.

4.5 Comparative Evaluation of DRL Algorithms

A systematic comparative evaluation was carried out to analyze the performance of PPO and TD3 in the learnt VAE-based surrogate environment. The objective was to determine the best policy optimization technique for sequential RUL estimation under the conditions of intricate time-dependent degradation dynamics. Differences in performance estimation can be attributable to different policy optimization strategies rather than differences in feature extraction because both algorithms employ the same learned latent representation. This experimental setup facilitates an objective comparison of the two algorithms' convergence behavior, trajectory prediction accuracy, and reward stability. The evaluation metrics stated in the preceding subsection were used to quantify performance, providing quantifiable evidence of each algorithm's respective strengths and limits for sequential RUL estimation. By conducting this analysis, the research gave provable answers on the effect of reward formulation and agent design on the learning dynamics, in addition to identifying the more suitable DRL algorithm for the proposed framework. This experimental design is intended to isolate how reward formulation and policy optimization strategy jointly shape sequential RUL estimation within a shared learned latent representation.

The modular architecture makes the framework readily adaptable across Prognostic Health Management (PHM) domains, even though it was tested on the CMAPSS turbofan dataset. Its modular design, which combines a distinct policy-learning module with a generative environment, makes it simple to adapt to systems that rely on high-dimensional sensor input and at the same time, lack reliable physical models. The method can be used for rotating equipment prediction, industrial machinery maintenance, EV battery RUL estimation, structural health monitoring, and many more. It works well in dynamic or data-scarce contexts where flexibility and generalization are crucial since it just needs historical run-to-failure data and uses learned latent dynamics with model-free reinforcement learning.

4.6 Experimental Setup

4.6.1 Data and Training Configuration

The NASA CMAPSS benchmark dataset's FD002 subset, which captures multivariate time-series data from turbofan engines running under different conditions and fault kinds, was used for the tests. Because of its complexity and realistic deterioration patterns, this dataset is especially well-suited for evaluating sequential RUL estimation frameworks under complex and heterogeneous degradation conditions. The dataset was partitioned into three disjoint subsets. Two hundred engine units were used to train the VAE surrogate environment, ten units were used for DRL policy optimization, and the remaining fifty units were reserved for out-of-sample evaluation to assess the generalization capability of the learned latent-space policies. Although only ten engine units were used for policy optimization, the reinforcement learning agent interacts repeatedly with the learned surrogate environment over multiple episodes, generating a substantially larger number of state-action transitions than the original trajectories alone. Consequently, this training setup provides sufficient experience for stable policy learning while maintaining a clear separation between surrogate training, policy optimization, and independent evaluation. To evaluate computational scalability, each model was trained and evaluated on CPU and GPU platforms. While predictive performance remained consistent across platforms, the use of an NVIDIA RTX 4090 GPU (24 GB VRAM) significantly accelerated training. To be specific, the time required to complete 300,000 PPO training steps was shortened from 60 minutes on CPU to 25 minutes on GPU, highlighting the substantial computational efficiency gains of GPU-accelerated training for sequential latent-space policy optimization. Also at this stage it is worth noting that the TD3 agent converged in 200,000 steps, but the PPO agent was trained for 300,000 steps. Based on convergence stability seen in early experiments, these training durations were chosen. The sample-efficient, off-policy architecture of TD3 contrasts with the on-policy nature of PPO, which usually requires more frequent policy changes, in terms of the necessary steps.

4.6.2 Surrogate Environment and Algorithm Configuration

CMAPSSEnv, a custom RL environment, was created in compliance with the OpenAI Gym interface. This environment allows the DRL agent to interact in a compact learned latent representation by integrating the learned VAE decoder as a differentiable surrogate simulator. The agent receives a compact latent representation of the system state, generated from sensor measurements and associated RUL information, and outputs a continuous latent action that guides transitions within the learned surrogate environment. The VAE decoder subsequently maps these latent transitions to reconstructed sensor observations and sequential RUL estimates. The Stable-Baselines3 package was used to implement the PPO and TD3 algorithms. Default hyperparameters, including the learning rate, discount factor, entropy coefficient, and batch size, were adopted for both algorithms to ensure a fair and consistent comparison. To keep the emphasis on methodological integrity rather than optimization bias, no algorithm-specific hyperparameter tuning was carried out. Reproducibility, interpretability, and modularity were prioritized in the experimental design. As described in Section 4, the experimental setting enabled controlled enabled a controlled comparison of the two policy optimization algorithms under identical surrogate dynam-

ics, latent representations, and reward formulations. The complete experimental workflow, from VAE surrogate training to latent-space policy optimization and final RUL evaluation, is summarized in Figure 4.

5 Results and Discussion

This section evaluates the proposed framework from three complementary perspectives: (i) the fidelity of the learned VAE-NARX surrogate environment, (ii) the effectiveness of latent-space policy optimization for sequential RUL estimation, and (iii) the comparative performance of PPO and TD3 under different reward formulations. Additional ablation experiments investigate on the effect of decoder architecture and latent dimensionality on downstream policy quality.

5.1 Validation of the VAE-NARX Surrogate Model

The VAE component was trained for 30 epochs using the Adam optimizer (learning rate = 0.0005) on 200 engine units (41,456 time-step samples). Training was evaluated against a held-out set of 50 engine units (10,271 samples) that were not used during VAE or RL training. Both the training and validation loss curves, shown in Figure 5(b), exhibit steady convergence, indicating stable learning as well as successful generalization.

To quantitatively assess reconstruction fidelity, Table 3 reports comprehensive reconstruction metrics on both training and evaluation sets. The VAE achieves an overall MSE of 0.001053 and R^2 of 0.9922 on the held-out evaluation set, confirming that the learned latent representation preserves the essential structure of the input data. The RUL-specific reconstruction MAE of 0.0396 on the evaluation set demonstrates that the surrogate environment accurately reproduces degradation trajectories with minimal information loss due to the

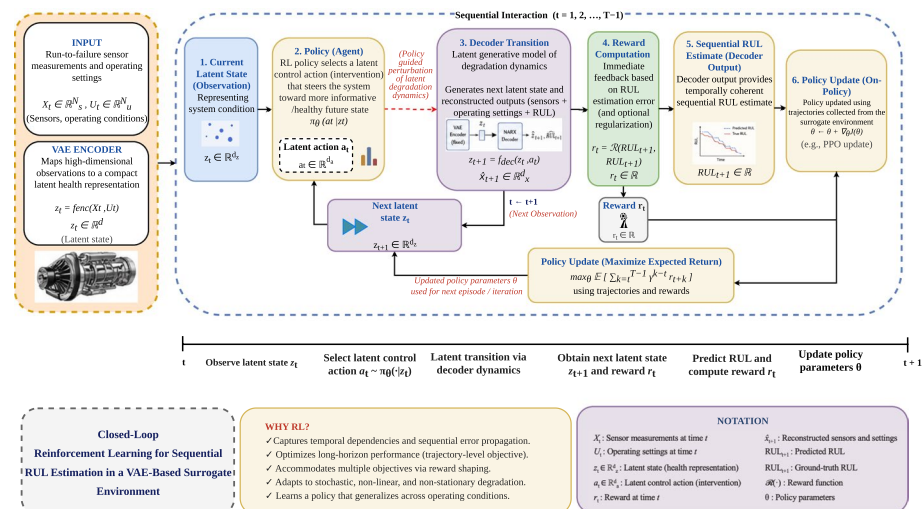


Fig. 4 Conceptual illustration of sequential latent-space policy optimization. At each timestep, the policy observes the current latent state, selects a latent action, receives a reward based on the resulting RUL estimate, and updates the policy through interaction with the VAE-based surrogate environment

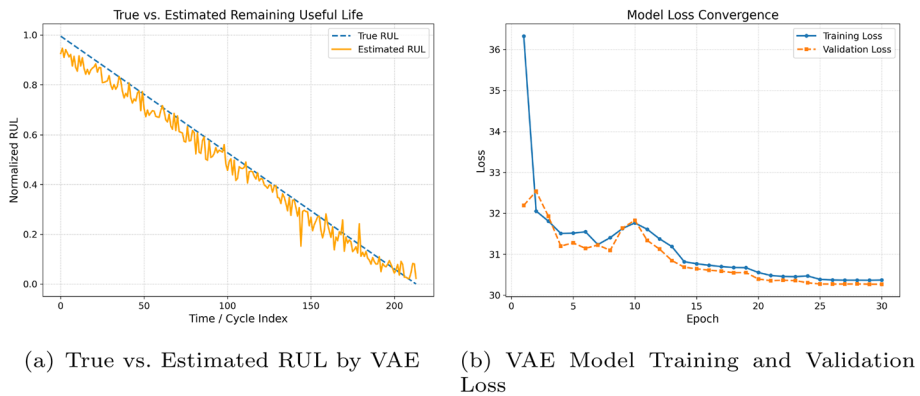


Fig. 5 Performance of the trained variational autoencoder (VAE)

Table 2 Reconstruction accuracy by degradation phase (evaluation set)

Degradation phase	Overall MSE	Overall MAE	RUL MSE	RUL MAE	KL Divergence
Early Life (RUL > 66%)	0.001080	0.021394	0.004588	0.048569	12.392
Mid Life (33%–66%)	0.000926	0.019842	0.003040	0.040929	12.177
Late Life (RUL ≤ 33%)	0.001151	0.020903	0.001481	0.029031	12.207

latent bottleneck. The KL divergence of 12.26 on the evaluation set, which nearly matches the training set value of 12.25, indicates that the latent space retains a well-regularized distribution while avoiding overfitting.

Table 2 splits down reconstruction quality by degradation stage, allowing for additional investigation across multiple operational phases. The findings show that RUL reconstruction error is highest in early life (MAE = 0.0486) and gradually declines from midlife (0.0409) to late life (0.0290). This pattern is consistent with the observation that early degradation phases have reduced signal variation, which makes precise state reconstruction more difficult. Importantly, reconstruction accuracy improves as equipment nears failure, which is when forecast accuracy is most important operationally. Figure 6 depicts the reconstruction quality patterns, including RUL correlation, phase-wise error distribution, and per-feature MSE breakdown.

These quantitative results demonstrate that the VAE-NARX surrogate accurately captures the underlying degradation dynamics and provides a reliable simulation environment for downstream reinforcement learning. The consistently low reconstruction error and high reconstruction fidelity indicate that prediction errors introduced by the surrogate remain sufficiently small to support stable downstream policy learning. These findings, together

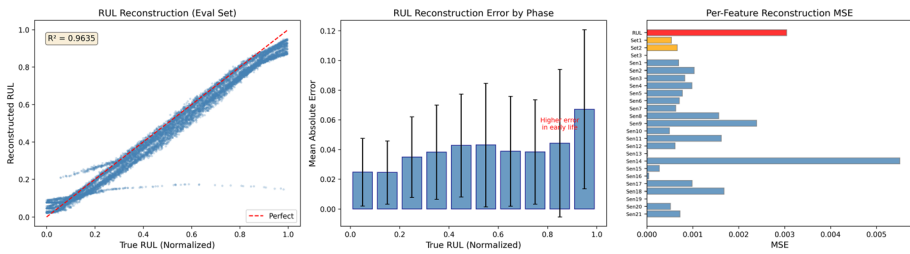


Fig. 6 Reconstruction quality analysis of the VAE model, including RUL correlation, phase-wise reconstruction error, and feature-level mean squared error

Table 3 VAE reconstruction quality on C-MAPSS FD002

Metric	Training set (200 engines)	Evaluation set (50 engines)
MSE	0.000836	0.001053
MAE	0.018738	0.020719
RMSE	0.028909	0.032446
R^2	0.9938	0.9922
KL Divergence	12.249	12.260
RUL-specific MSE	0.002549	0.003045
RUL-specific MAE	0.036000	0.039555
Sensor features MSE	0.000816	0.001051
Operational settings MSE	0.000405	0.000400

with the qualitative trajectory comparisons shown in Figure 5, justify the use of the learned surrogate environment for sequential latent-space policy optimization (Table 3).

5.1.1 Ablation Study: Decoder Architecture

An ablation study was carried out to support the selection of the NARX-based decoder architecture by contrasting it with a conventional feedforward MLP decoder with comparable parameter counts (47,897 decoder parameters in both cases). The encoder architectures utilized in both variations were the same, and they were trained using the same parameters (30 epochs, Adam optimizer, learning rate 0.0005).

The results are shown in Table 4. Reconstruction quality is significantly improved by the NARX decoder in all criteria. The R^2 increases from 0.9587 to 0.9876, while the overall MSE decreases by 70.0% (0.001679 vs. 0.005591). The biggest difference may be observed in RUL-specific reconstruction, where the NARX decoder achieves an RUL MSE of 0.0153, which is 5.1 times better than the standard decoder's 0.0777. This indicates that the temporal dependencies present in degradation trajectories, which a simple feedforward mapping is unable to capture, are best captured by the autoregressive structure of the NARX decoder.

Additionally, the standard decoder shows a much higher KL divergence (622,302 vs. 17.09), indicating that in the absence of the structured decoding pathway, the encoder-decoder pair is unable to learn a well-regularized latent space. The training convergence curves and reconstruction error comparison between the two decoder architectures are shown in Figure 7. These findings support the choice of the NARX-based architecture for

Table 4 Ablation study, NARX vs standard MLP decoder (evaluation set)

Metric	NARX decoder (proposed)	Standard MLP decoder
Overall MSE	0.001679	0.005591
Overall MAE	0.023626	0.036864
R^2	0.9876	0.9587
RUL MSE	0.015319	0.077719
RUL MAE	0.076052	0.238554
KL divergence	17.09	622,302
Final training loss	30.43	31.59
Final validation loss	30.34	31.36
Decoder parameters	47,897	47,897

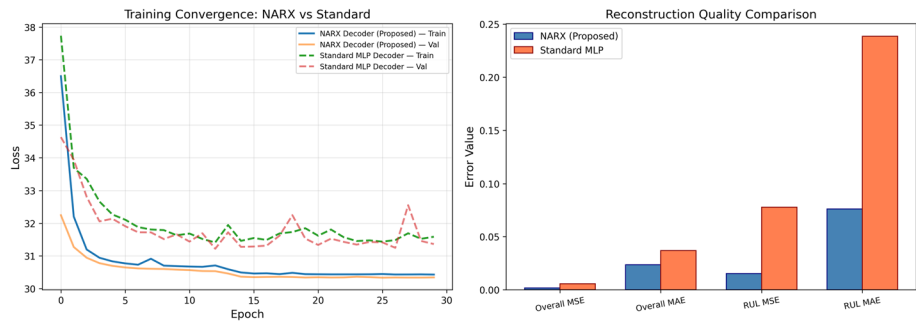


Fig. 7 Ablation study comparing the proposed NARX decoder with a standard MLP decoder in terms of training convergence and reconstruction error metrics

downstream RUL estimation techniques by demonstrating that it offers significant gains in surrogate environment fidelity.

These results show that the decoder’s capacity to maintain degradation dynamics necessary for downstream policy optimization influences surrogate environment fidelity in addition to reconstruction accuracy.

5.2 Comparative Performance of PPO and TD3 for RUL Estimation

A systematic comparative analysis was carried out to evaluate the performance of PPO and TD3 under twelve reward formulations after the VAE-NARX surrogate environment was validated. The comparison looks at resilience, learning stability, reward convergence, and prediction accuracy across several optimization objectives.

5.2.1 A Graphical Comparison

Smoother reward trajectories and faster PPO convergence are shown in initial trials with single-component loss functions like as Mean Squared Error (MSE), Huber, and Logcosh. While the corresponding TD3 trajectories [Figure 8 (d, e, f)] show greater variance under identical reward formulations, indicating a different sensitivity of the two optimization mechanisms to the surrogate’s reward landscape.

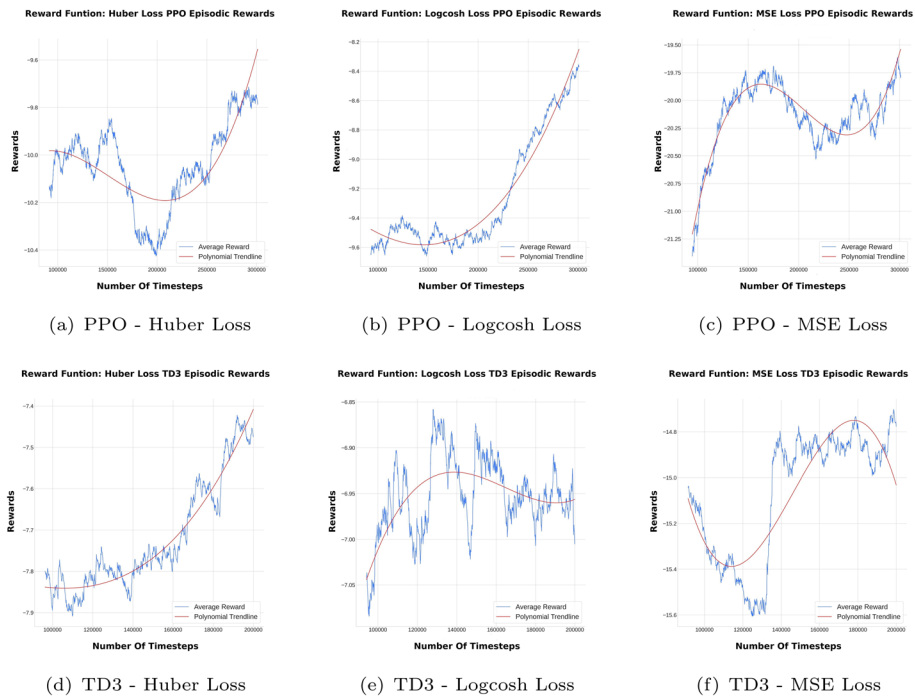


Fig. 8 Episodic rewards for PPO and TD3 with single-component losses

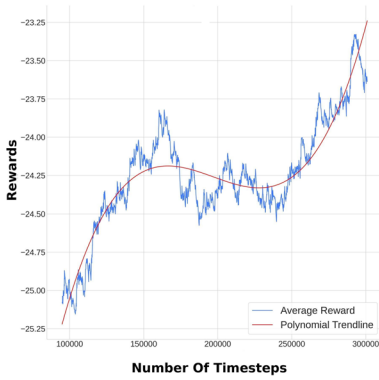
Figure 8 (a, b, c) show that single-component loss formulations support smoother reward convergence within the learned latent environment. When considered collectively, this aggregate comparison across single-loss reward functions indicates that a policy optimization mechanism's reliability in converging within the learnt latent environment is significantly impacted by the smoothness of the underlying reward landscape.

Hybrid reward formulations produced additional gains in learning stability and cumulative reward for both agents. The richer feedback signal from composite rewards such as Huber + MAE and MAE + Logcosh translated into smoother convergence under the on-policy formulation [Figure 9 (a, b)], while the corresponding off-policy trajectories [Figure 9 (c, d)] remained comparatively noisy. This asymmetry is consistent with the clipped surrogate objective providing more effective utilization of multi-component reward structures than value-based bootstrapping in this latent environment.

On the other hand, both agents faced considerable difficulties with reward systems that are extremely sensitive to outliers, including MAPE and cosine similarity-based losses. Under these formulations, PPO demonstrated elevated reward variance and unstable learning behavior [Figure 10 (a, b, c)], whereas TD3 demonstrated significant instability and, in multiple instances, exhibited unstable convergence characterized by large reward oscillations [Figure 10 (d, e, f)]. These patterns indicate that these reward systems are less appropriate for sequential RUL estimation in learned, non-stationary contexts because they offer noisy or poorly scaled gradients.

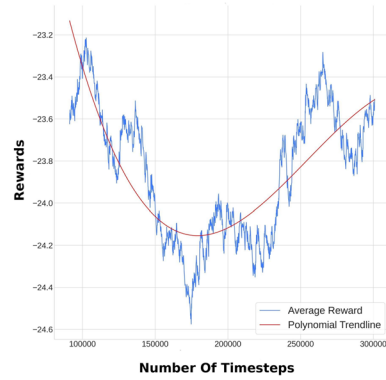
Qualitative evaluation of RUL prediction trajectories was performed on representative test systems alongside the episodic reward analysis. PPO and TD3 predictions for Test Sys-

Reward Functon: Combination Of Huber and MAE Loss functions PPO Episodic Rewards



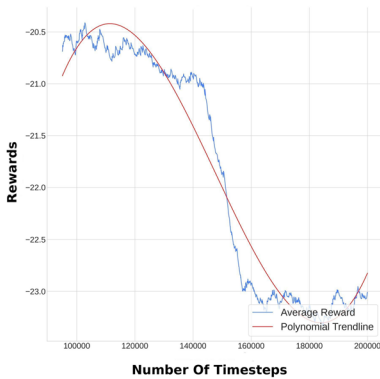
(a) PPO - Huber + MAE

Reward Functon: Combination Of MAE and Logcosh Loss functions PPO Episodic Rewards



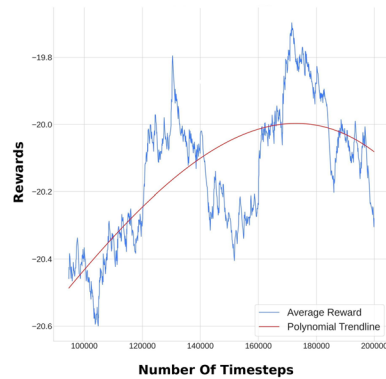
(b) PPO - MAE + Logcosh

Reward Functon: Combination Of Huber and MAE Loss functions TD3 Episodic Rewards



(c) TD3 - Huber + MAE

Reward Functon: Combination Of MAE and Logcosh Loss functions TD3 Episodic Rewards



(d) TD3 - MAE + Logcosh

Fig. 9 Episodic rewards for PPO and TD3 with hybrid reward functions

tems 31 and 34 are shown in Figures 11 and 12. The two methods differ most in the late-cycle regime: the on-policy agent maintains smoother, largely monotonic trajectories that closely follow the true degradation trend, whereas the off-policy agent is more sensitive to stochastic late-cycle dynamics, reflecting differences in value propagation under non-stationary degradation.

Figure 13 illustrates a similar behavior for randomly selected test instances (Systems 5 and 18). The temporal consistency of the estimated trajectories differed depending on when the projection started within the degradation lifespan. In the on-policy formulation, trajectories were consistent even when they began early in the operating life (10–16%), while the off-policy formulation was more sensitive to short-term fluctuations, showing differences in robustness and resilience for real-time PdM deployment.

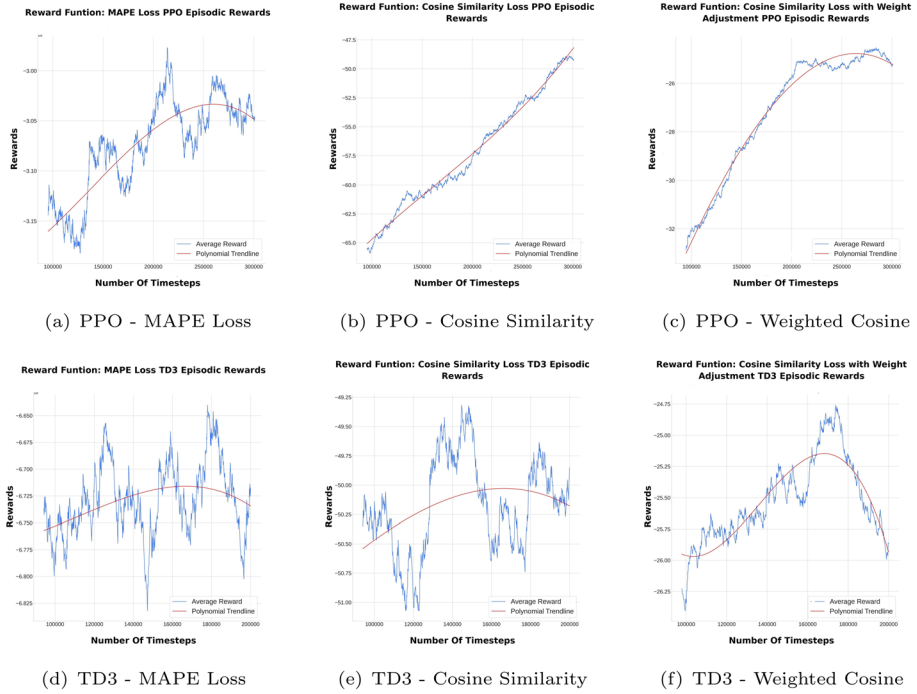


Fig. 10 Episodic rewards for PPO and TD3 with outlier-sensitive reward functions

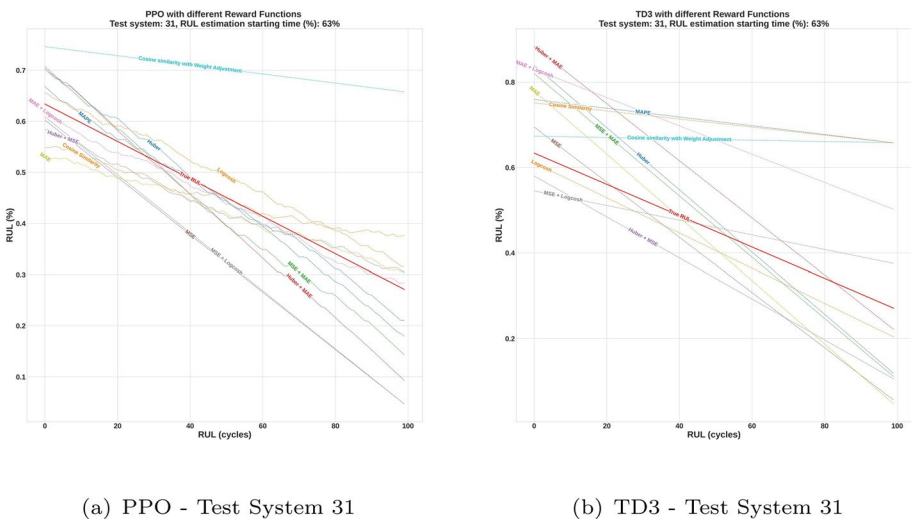


Fig. 11 Comparison of PPO and TD3 RUL estimates for test system 31

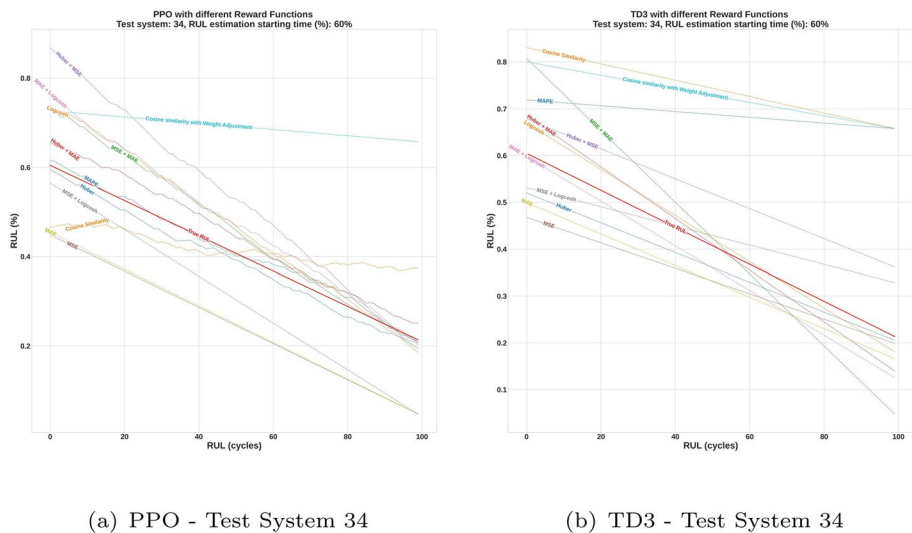


Fig. 12 Comparison of PPO and TD3 RUL estimates for Test System 34

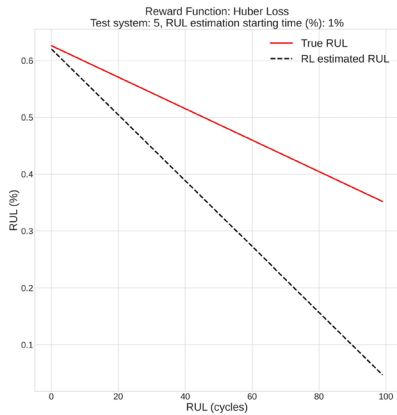
Figure 14 illustrates predictions under the top-performing reward formulations (MAPE, MAE, and the hybrid MAE + Logcosh). Forecast stability is closely tied to both reward choice and degradation stage. Accuracy remains high across mid-life and late-stage scenarios (49–88%), with the MAE + Logcosh formulation producing especially stable behavior near critical failure thresholds.

This encourages the use of hybrid, robustness-oriented reward structuring for long-horizon RUL estimation, when late-stage reliability is crucial.

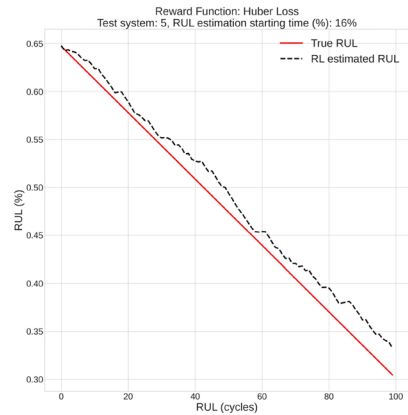
5.2.2 A Quantitative Comparison

The performance of the PPO and TD3 agents was systematically assessed across twelve reward formulations using three evaluation seeds (42, 124, 777) to quantify result variability. Table 5 presents the mean episodic reward with standard deviations, along with Cohen's d effect sizes computed on RMSE values to measure the practical magnitude of performance differences.

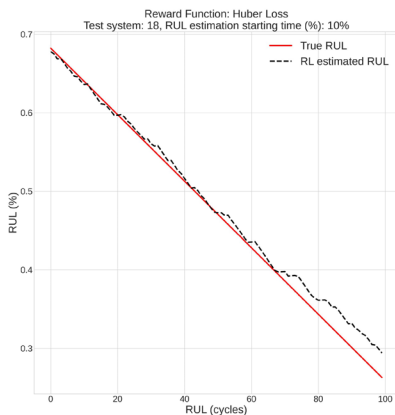
Across the twelve reward formulations, the on-policy agent achieves lower RMSE than the off-policy agent in 11 of 12 configurations (Table 6). Cohen's d effect sizes exceed the large-effect threshold ($|d| > 0.8$) in 10 of these 12 configurations, indicating that the differences in downstream prediction accuracy are consistent and practically meaningful rather than incidental to a particular reward choice. The best-performing configuration, PPO with MSE + MAE Combination reward, achieves an RMSE of 0.1262 ± 0.0040 , MAE of 0.0963 ± 0.0054 , and R^2 of 0.7257 ± 0.0154 , representing the lowest prediction error and highest explained variance across all configurations. Under the Huber Loss, PPO achieves an RMSE of 0.1512 ± 0.0193 compared to 0.2161 ± 0.0046 for TD3, with a large effect size of $d = 4.64$. The hybrid Huber + MAE Combination yields an RMSE of 0.1515 ± 0.0170 for PPO versus 0.3490 ± 0.0074 for TD3, with Cohen's $d = 15.07$, among the largest observed differences.



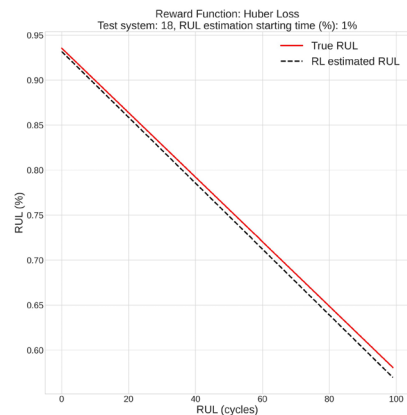
(a) TD3 - System 5 (start: 1%)



(b) PPO - System 5 (start: 16%)



(c) PPO - System 18 (start: 10%)

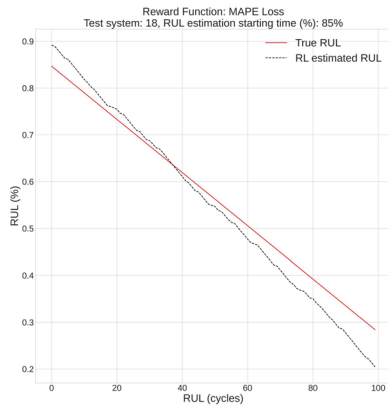


(d) TD3 - System 18 (start: 1%)

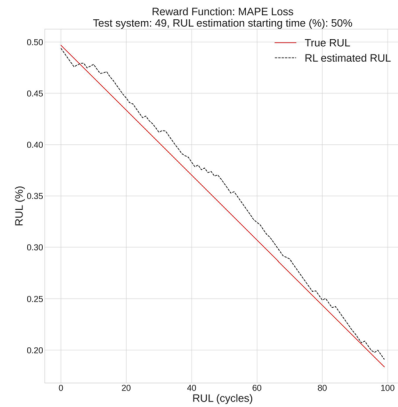
Fig. 13 Qualitative RUL prediction comparison on randomly selected test systems using Huber Loss

The MAPE reward function exhibits anomalous behavior with rewards on the order of 10^8 to 10^9 (Table 5). This instability arises from its formulation $(100/n) \times \sum |y - \hat{y}|/|y|$, where the denominator approaches zero as the true NormTime nears end-of-life (50 out of 10,271 samples, 0.5%), causing divergence. Despite this, PPO trained with MAPE achieves a competitive RMSE of 0.1338 ± 0.0080 (Table 6), indicating that useful representations are learned before reward degradation. However, MAPE remains unsuitable for RUL estimation tasks where the target approaches zero.

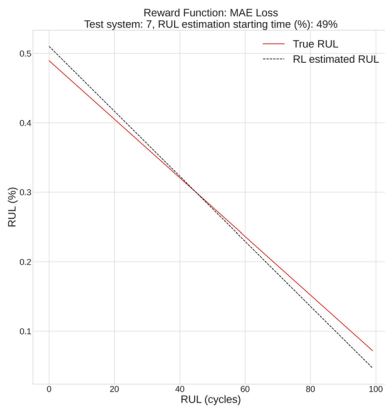
Similarly, the Cosine Similarity and Weighted Cosine Similarity reward functions produce elevated RMSE values (0.32–0.35) for both algorithms, indicating that directional similarity metrics provide insufficient gradient information for effective policy learning in degradation-aware sequential RUL estimation.



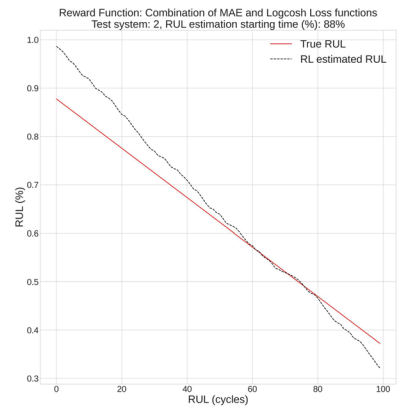
(a) MAPE Loss - System 18 (Start: 85%)



(b) MAPE Loss - System 49 (Start: 50%)



(c) MAE Loss - System 7 (Start: 49%)



(d) MAE + Logcosh - System 2 (Start: 88%)

Fig. 14 PPO-based RUL predictions under top-performing reward functions across diverse degradation trajectories

To validate the proposed framework against conventional approaches, Table 7 compares the best PPO and TD3 configurations with supervised learning baselines trained on the same features and evaluation split. PPO with the MSE + MAE Combination reward achieves an RMSE of 0.1262, representing a 26.7% improvement over Linear Regression (0.1721) and 40.3% over LSTM (0.2114). This result demonstrates that the proposed VAE-DRL framework provides tangible predictive advantages over conventional supervised learning approaches. Unlike the supervised baselines, which directly learn a mapping from sensor observations to RUL values, the proposed framework optimizes sequential latent trajectories through interaction with a learned VAE-NARX surrogate environment. This enables temporally coherent RUL estimation by explicitly modeling degradation evolution during policy optimization rather than relying solely on point-wise prediction.

Table 5 Mean episodic reward ($\pm$ standard deviation) for TD3 and PPO across twelve reward formulations. Cohen's d effect sizes, computed on RMSE values across three evaluation seeds, quantify the practical magnitude of prediction accuracy differences; $|d| > 0.8$ indicates a large effect. Standard RUL evaluation metrics (RMSE, MAE, R^2) are reported separately in Table 6

Sr. No.	Reward function	TD3 (Mean $\pm$ Std)	PPO (Mean $\pm$ Std)	Cohen's d
1	Huber loss	6.05 $\pm$ 4.08	15.28 $\pm$ 5.81	+4.64 (Large)
2	Logcosh loss	10.05 $\pm$ 9.10	8.83 $\pm$ 4.38	+16.37 (Large)
3	Cosine similarity loss	53.45 $\pm$ 27.73	53.77 $\pm$ 38.87	+19.79 (Large)
4	MSE loss	13.54 $\pm$ 7.69	25.36 $\pm$ 12.02	+0.07 (Negl.)
5	MAPE loss (outlier)	7.59 $\times 10^8 \pm$ 4.40 $\times 10^8$	4.14 $\times 10^9 \pm$ 2.71 $\times 10^9$	+36.86 (Large)
6	MAE Loss	43.65 $\pm$ 17.42	40.47 $\pm$ 27.17	+3.30 (Large)
7	MSE + MAE combination	24.71 $\pm$ 14.18	29.97 $\pm$ 13.72	+19.67 (Large)
8	Huber + MAE combination	18.40 $\pm$ 9.83	32.69 $\pm$ 17.78	+15.07 (Large)
9	Huber + MSE combination	11.21 $\pm$ 7.57	16.77 $\pm$ 12.49	+4.25 (Large)
10	MAE + Logcosh combination	16.22 $\pm$ 8.41	21.25 $\pm$ 15.10	+5.00 (Large)
11	MSE + Logcosh combination	10.42 $\pm$ 7.62	15.24 $\pm$ 10.31	+7.08 (Large)
12	Weighted cosine similarity	22.19 $\pm$ 15.86	22.17 $\pm$ 13.42	+0.02 (Negl.)

Table 6 Standard RUL evaluation metrics for PPO and TD3 across all reward formulations. Results report mean $\pm$ standard deviation over three evaluation seeds (42, 124, 777). Best result per reward function is shown in *italics*

Sr. No.	Reward function	Algorithm	RMSE	MAE	R^2
1	Huber loss	PPO	<i>0.1512 $\pm$ 0.0193</i>	<i>0.1100 $\pm$ 0.0147</i>	<i>0.6058 $\pm$ 0.0778</i>
		TD3	0.2161 $\pm$ 0.0046	0.1568 $\pm$ 0.0059	0.1947 $\pm$ 0.0680
2	Logcosh loss	PPO	<i>0.1578 $\pm$ 0.0011</i>	<i>0.1252 $\pm$ 0.0045</i>	<i>0.5714 $\pm$ 0.0243</i>
		TD3	0.2381 $\pm$ 0.0069	0.1775 $\pm$ 0.0022	0.0211 $\pm$ 0.1099
3	Cosine similarity loss	PPO	<i>0.3172 $\pm$ 0.0017</i>	<i>0.2555 $\pm$ 0.0024</i>	<i>-0.7320 $\pm$ 0.0866</i>
		TD3	0.3459 $\pm$ 0.0012	0.2866 $\pm$ 0.0023	-1.0616 $\pm$ 0.1281
4	MSE loss	PPO	<i>0.1750 $\pm$ 0.0165</i>	<i>0.1257 $\pm$ 0.0176</i>	<i>0.4730 $\pm$ 0.0724</i>
		TD3	0.1759 $\pm$ 0.0087	0.1331 $\pm$ 0.0130	0.4681 $\pm$ 0.0250
5	MAPE loss	PPO	<i>0.1338 $\pm$ 0.0080</i>	<i>0.1064 $\pm$ 0.0067</i>	<i>0.6911 $\pm$ 0.0383</i>
		TD3	0.3459 $\pm$ 0.0012	0.2866 $\pm$ 0.0023	-1.0616 $\pm$ 0.1281
6	MAE loss	PPO	<i>0.1661 $\pm$ 0.0193</i>	<i>0.1203 $\pm$ 0.0191</i>	<i>0.5244 $\pm$ 0.0841</i>
		TD3	0.2136 $\pm$ 0.0063	0.1592 $\pm$ 0.0063	0.2131 $\pm$ 0.0791
7	MSE + MAE combination	PPO	<i>0.1262 $\pm$ 0.0040</i>	<i>0.0963 $\pm$ 0.0054</i>	<i>0.7257 $\pm$ 0.0154</i>
		TD3	0.2175 $\pm$ 0.0052	0.1648 $\pm$ 0.0069	0.1841 $\pm$ 0.0803
8	Huber + MAE combination	PPO	<i>0.1515 $\pm$ 0.0170</i>	<i>0.1117 $\pm$ 0.0163</i>	<i>0.6047 $\pm$ 0.0680</i>
		TD3	0.3490 $\pm$ 0.0074	0.2714 $\pm$ 0.0098	-1.0977 $\pm$ 0.1256
9	Huber + MSE combination	PPO	<i>0.1528 $\pm$ 0.0123</i>	<i>0.1152 $\pm$ 0.0103</i>	<i>0.5984 $\pm$ 0.0453</i>
		TD3	0.2045 $\pm$ 0.0120	0.1543 $\pm$ 0.0103	0.2804 $\pm$ 0.0608
10	MAE + Logcosh combination	PPO	<i>0.1498 $\pm$ 0.0195</i>	<i>0.1087 $\pm$ 0.0154</i>	<i>0.6130 $\pm$ 0.0790</i>
		TD3	0.2202 $\pm$ 0.0041	0.1603 $\pm$ 0.0062	0.1654 $\pm$ 0.0347
11	MSE + Logcosh combination	PPO	<i>0.1690 $\pm$ 0.0146</i>	<i>0.1198 $\pm$ 0.0161</i>	<i>0.5089 $\pm$ 0.0590</i>
		TD3	0.2909 $\pm$ 0.0195	0.2207 $\pm$ 0.0173	-0.4570 $\pm$ 0.1518
12	Weighted Cosine similarity	PPO	<i>0.3459 $\pm$ 0.0012</i>	<i>0.2866 $\pm$ 0.0023</i>	<i>-1.0613 $\pm$ 0.1282</i>
		TD3	0.3459 $\pm$ 0.0012	0.2866 $\pm$ 0.0023	-1.0616 $\pm$ 0.1281

Table 7 Comparison of the proposed VAE-DRL framework with supervised learning baselines on the C-MAPSS FD002 evaluation set (50 engine units). All methods use the same input features and train/evaluation split

Method	RMSE	MAE
<i>Supervised Baselines</i>		
Linear Regression	0.1721	0.1382
MLP (64, 64)	0.1766	0.1383
MLP (256, 128, 64)	0.1815	0.1366
LSTM (64-32)	0.2114	0.1629
<i>Proposed VAE-DRL Framework</i>		
PPO (MSE + MAE Combination)	<i>0.1262 ± 0.0040</i>	<i>0.0963 ± 0.0054</i>
TD3 (MSE Loss)	0.1759 ± 0.0087	0.1331 ± 0.0130

Table 8 Latent dimension ablation: VAE surrogate reconstruction quality (held-out evaluation set) and PPO policy performance across all twelve reward formulations. Dim = 1 is the proposed configuration; results for Dim = 8, 16, and 32 are from newly trained VAEs evaluated under identical conditions. PPO is used consistently as the proposed algorithm across all dimensions. Best PPO RMSE per dimension is shown in *italics*

Latent Dim	VAE Surrogate Quality (Eval Set)			PPO Downstream RL	
	R ²	RUL MSE	KL Div	Best RMSE	Mean RMSE
1 (proposed)	0.9922	0.003045	12.26	<i>0.1262 ± 0.0040</i>	0.1830
8	0.9922	0.001422	12.35	0.2609 ± 0.0120	0.3442
16	0.9928	0.001534	12.66	0.1349 ± 0.0026	0.1498
32	0.9932	0.002801	14.48	0.1879 ± 0.0067	0.2481

5.2.3 Ablation Study: Effect of Latent Dimension on Surrogate Fidelity and Downstream Policy Quality

Having established PPO as the proposed algorithm in Section 5, an ablation study was conducted to directly address how surrogate modeling choices propagate into downstream policy learning outcomes. The VAE latent dimension was varied across four configurations: 1, 8, 16, and 32. For each configuration, the VAE was trained to full convergence under identical conditions (30 epochs, Adam optimizer, learning rate 0.0005), and the resulting surrogate was used to train a PPO agent across all twelve reward formulations. Table 8 reports VAE reconstruction quality on the held-out evaluation set alongside the best and mean PPO RMSE per dimension.

A key finding of this ablation study is that surrogate reconstruction quality alone is not a reliable predictor of downstream policy performance. Dim = 1 achieves the best downstream PPO RMSE of 0.1262 ± 0.0040 and the most consistent policy learning across reward formulations: PPO outperforms TD3 in 11 out of 12 reward configurations at Dim = 1, compared to only 3 out of 12 at Dim = 8 and 7 out of 12 at Dim = 32. This consistency is a stronger indicator of robust policy learning than best-case RMSE alone. Dim = 8, despite achieving the lowest VAE RUL-specific reconstruction MSE (0.001422) among all configurations, yields the worst mean PPO RMSE (0.3442) and best PPO RMSE of only 0.2609 ± 0.0120 — a 106.7% increase relative to Dim = 1. This indicates that lower reconstruction error does not guarantee better downstream policy performance. Dim = 16 achieves competitive performance (best PPO RMSE = 0.1349 ± 0.0026 , +6.9% vs Dim = 1) with consistent PPO superiority across all 12 reward configurations, making it the strongest

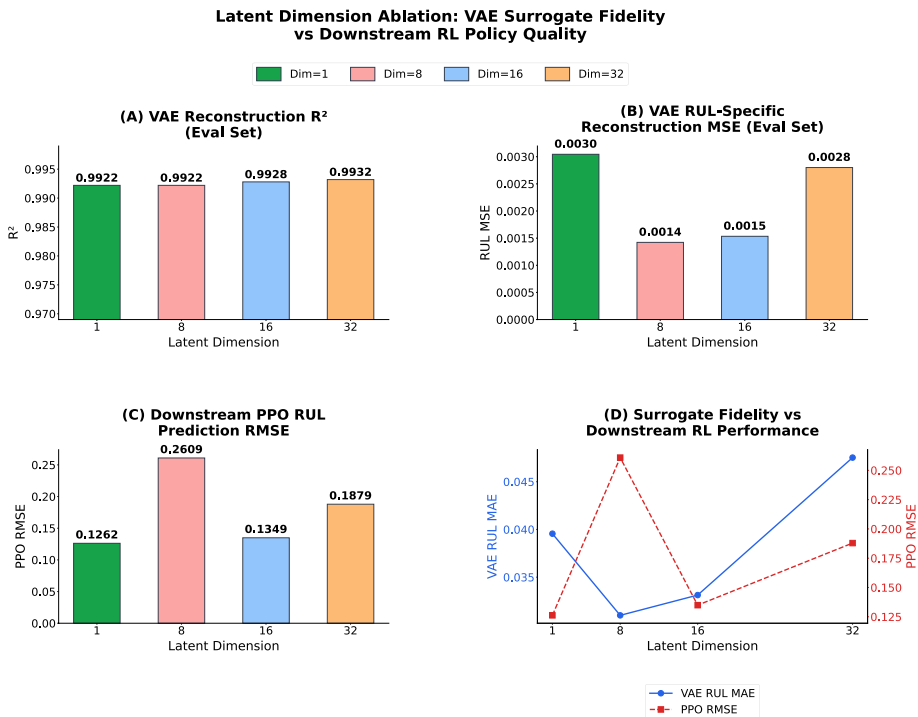


Fig. 15 Latent dimension ablation: PPO policy performance (downstream RUL prediction RMSE) across four latent dimensions (1, 8, 16, 32), evaluated under all twelve reward formulations. The mean PPO RMSE across all rewards (dashed line) shows that Dim = 1 and Dim = 16 achieve the lowest and most consistent performance, while Dim = 8 and Dim = 32 show substantially higher error. This illustrates that latent dimensionality critically governs downstream policy learning stability, independently of VAE reconstruction quality

alternative to the proposed Dim = 1 configuration. Dim = 32 degrades further (best PPO RMSE = 0.1879 ± 0.0067 , +48.9% vs Dim = 1).

These findings suggest that the one-dimensional latent bottleneck provides a compact representation that is more amenable to stable policy optimization across diverse reward formulations. Higher-dimensional latent spaces, while achieving lower reconstruction error, introduce a more complex optimization landscape in which PPO policy learning becomes less stable and consistent. The choice of latent dimensionality is therefore a critical design decision whose effects on policy quality cannot be predicted from VAE reconstruction metrics alone. Figure 15 illustrates the PPO RMSE trends across all four configurations and twelve reward formulations. These findings further demonstrate that the effectiveness of the proposed framework depends not only on the fidelity of the learned surrogate environment but also on how effectively its latent representation supports downstream policy optimization for sequential RUL estimation. More broadly, these findings suggest that surrogate models for reinforcement learning should be evaluated not only by reconstruction quality but also by their ability to support effective downstream policy optimization.

5.3 Discussion

The quantitative validation of the VAE surrogate environment provides crucial context for interpreting reinforcement learning outcomes. The phase-wise analysis reveals that VAE reconstruction accuracy is highest during the late degradation stage (RUL MAE = 0.029), where accurate sequential RUL estimation is most critical for supporting maintenance planning. Although reconstruction error is relatively higher during the early degradation stage, accurate modeling of the late degradation phase is particularly important because prediction uncertainty typically has the greatest operational impact as failure approaches. Furthermore, the ablation study confirms that the proposed NARX decoder contributes substantially to surrogate environment fidelity, achieving a $5.1\times$ improvement in RUL reconstruction accuracy compared with a standard feedforward decoder. The latent-dimension ablation (Section 5.2.2) further demonstrates that surrogate modeling choices propagate into downstream policy quality in a non-monotonic manner. Although Dim = 8 achieves the lowest VAE RUL reconstruction error, it produces the largest degradation in PPO performance, confirming that reconstruction accuracy alone is not a reliable predictor of downstream policy learning performance. This enhanced surrogate fidelity likely contributes to the stable policy optimization observed with PPO by providing more accurate latent state transitions during training.

The quantitative evaluation using standard RUL metrics (RMSE and MAE) together with Cohen's d effect size analysis provides stronger statistical evidence than mean reward comparisons alone. The consistently large effect sizes observed in 10 of the 12 reward configurations suggest that PPO's on-policy optimization strategy is better suited to the non-stationary degradation dynamics modeled by the VAE-NARX surrogate environment. Notably, the two configurations where Cohen's d is negligible, namely MSE Loss ($d = 0.07$) and Weighted Cosine Similarity ($d = 0.02$), correspond to cases where either both algorithms perform similarly well (MSE Loss) or both fail to learn effective policies (Weighted Cosine Similarity), further supporting the discriminative value of the statistical analysis.

The comparison with supervised baselines (Table 7) addresses the necessity of the reinforcement learning formulation. The best PPO configuration outperforms all evaluated supervised methods, including an LSTM with comparable architectural complexity. This improvement can be attributed to two complementary factors. First, unlike supervised approaches that directly regress RUL from sensor observations, the proposed framework optimizes latent degradation trajectories through interaction with the learned surrogate environment, resulting in more temporally coherent sequential RUL estimates. Second, reward shaping provides a flexible optimization mechanism that allows the policy to balance multiple aspects of prediction quality through composite reward formulations such as MSE + MAE and Huber + MAE.

The anomalous performance observed with the MAPE reward provides an important practical insight for reward engineering in reinforcement learning-based prognostics. Reward functions involving division by the target variable are fundamentally unsuitable for RUL estimation because the target approaches zero near failure, resulting in numerical instability and poor optimization behavior. These findings provide practical guidance for designing reward formulations and further demonstrate that reward design should be regarded as a fundamental component of reinforcement learning-based prognostics rather than merely a hyperparameter tuning strategy.

The observed performance advantage of PPO may be attributed to its on-policy optimization strategy, which continuously updates the policy using trajectories generated by the latest interactions with the VAE-NARX surrogate environment. Such an optimization mechanism enables PPO to adapt more effectively to the non-linear and stochastic degradation dynamics represented within the learned latent space while maintaining stable policy updates throughout training.

The comparatively lower performance of TD3 may be attributed to the challenges of accurately estimating the value function within the learned latent-space representation. As an off-policy algorithm, TD3 relies on bootstrapped value targets, which may become more sensitive to approximation errors when interacting with a non-stationary surrogate environment. Consequently, long-horizon RUL estimation can become more susceptible to accumulated value estimation errors in the presence of complex degradation dynamics and noisy latent transitions. These observations suggest that, under the experimental conditions considered in this study, off-policy value-based methods may be more sensitive to surrogate modeling errors than on-policy policy optimization approaches such as PPO.

This analysis provides two important insights. First, the experimental results show that for sequential RUL estimation in the proposed surrogate environment, the on-policy actor-critic strategy represented by PPO is more successful than the assessed off-policy alternative. This finding provides empirical support for algorithm selection in reinforcement learning-based prognostics and validates the application of policy-based optimization for intricate degradation processes.

Second, the methodical assessment of twelve reward formulations shows that a reward design is essential to prognostics based on reinforcement learning. Optimization stability, convergence behavior, and downstream prediction performance are consistently enhanced by hybrid loss-based reward formulations, especially those that combine robustness and prediction accuracy. These results show that reward shaping is not merely a hyperparameter tuning technique but also an essential design element of effective DRL-based RUL estimation.

An important implication of our research is that surrogate models intended for reinforcement learning should not be evaluated exclusively using reconstruction accuracy. The latent-dimension ablation demonstrates that latent representations producing lower reconstruction error may nevertheless yield inferior downstream policy optimization performance. Consequently, surrogate-assisted reinforcement learning frameworks should jointly evaluate surrogate fidelity and downstream policy learning effectiveness to identify latent representations that are both accurate and optimization-friendly.

Despite these contributions, The VAE-based surrogate environment may introduce reconstruction errors, particularly during the early degradation stage where sensor variations are limited. Moreover, the proposed methodology has been validated on the FD002 subset of the NASA C-MAPSS benchmark and hence requires further validation on larger PHM datasets featuring more complex degradation behaviors, multi-component systems, and concurrent failure scenarios. Finally, the proposed methodology does not explicitly quantify predictive uncertainty, which may limit its applicability in safety-critical contexts where reliability and interpretability are essential. Future study will investigate uncertainty-aware reinforcement learning using Bayesian latent representations, ensemble policy learning, probabilistic policy optimization, online continual learning, and hierarchical reinforcement learning for large-scale industrial prognostics.

Overall, the proposed VAE-DRL framework shows that surrogate-assisted reinforcement learning is a versatile and scalable solution to sequential RUL estimation in data-driven PHM applications. By combining learnt latent representations with policy optimization, the approach eliminates the requirement for manually constructed degradation models while maintaining high predictive performance. These properties make it a suitable starting point for future intelligent prognostics and predictive maintenance solutions.

6 Conclusion and Future Work

This paper introduces a VAE-NARX-based Deep Reinforcement Learning (DRL) framework for sequential Remaining Useful Life (RUL) estimate, which combines a learnt surrogate environment with latent-space policy optimization. In the proposed framework, a Variational Autoencoder (VAE) compresses high-dimensional sensor observations into compact latent representations, and a NARX-based decoder analyzes degradation dynamics using a differentiable surrogate environment. Quantitative validation supports the surrogate model's fidelity, with a reconstruction R^2 of 0.9922 on the held-out evaluation set. The decoder ablation study shows a 5.1× improvement in RUL reconstruction accuracy compared to a standard feedforward decoder, indicating that the learnt surrogate environment is a reliable foundation for downstream reinforcement learning.

The experimental evaluation on the NASA C-MAPSS FD002 benchmark demonstrates that both the policy optimization algorithm and the reward formulation are crucial for determining learning stability and sequential prediction performance. Across almost all twelve reward formulations, the results show that these two design choices are closely associated in surrogate-assisted sequential RUL estimation. The MSE + MAE reward formulation outperformed all other settings, with an RMSE of 0.1262 ± 0.0040 and an MAE of 0.0963 ± 0.0054 . The proposed approach also consistently outperformed traditional supervised learning baselines, such as linear regression and LSTM models. In addition, a systematic comparison of twelve reward formulations revealed that hybrid reward functions improve optimization stability and prediction accuracy, whereas outlier-sensitive objectives like MAPE and cosine similarity become numerically unstable as target values approach zero. These findings offer useful recommendations for constructing reward functions in reinforcement learning-based prognostics.

The methodological contribution of this study goes beyond predictive performance. The latent-dimension ablation analysis demonstrates that improving VAE reconstruction quality alone does not always result in better downstream policy optimization. Instead, successful sequential RUL estimation relies on learning latent representations that retain degradation dynamics while simultaneously encouraging stable reinforcement learning. This study shows that surrogate environments should be evaluated not only based on reconstruction accuracy, but also on their potential to support effective policy learning.

Overall, the proposed framework enables temporally coherent sequential estimation without requiring manually constructed health indicators or explicit degradation models. Because of its modular design, the framework can be adapted to a wide range of and Health Management (PHM) applications that require high-dimensional sensor data, such as industrial machinery, wind turbines, aviation systems, robotics, and battery health monitoring.

Future work will focus on extending the framework to multi-component and interacting systems, exploring online and continual learning for streaming sensor data, and incorporating uncertainty-aware reinforcement learning through Bayesian latent representations or ensemble policy learning. Another promising direction is to translate the learned latent-space policies into physically interpretable maintenance recommendations while validating the framework across a broader range of industrial PHM benchmarks.

Acknowledgements We thank Gujarat Technological University and Bacancy DataProphets Private Limited for their support in this research.

Author Contributions HS, AM, and HI contributed to the conceptualization and implementation of the study and carried out data collection and analysis. HS drafted the main manuscript text. VDP and AP provided critical review and constructive feedback. The overall study was supervised by VDP. All authors reviewed and approved the final manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors state that they have no known financial or personal conflicting interests that could have appeared to influence the work reported in this paper.

Ethical approval This study does not include human participants, animals, or sensitive personal data. Therefore, ethics approval and agreement to participate were not necessary.

Consent for publication All authors have reviewed and approved the final manuscript and provided their consent for publication.

References

- Al-Dulaimi, A., Zabihi, S., Asif, A., et al. (2019). Hybrid deep neural network model for remaining useful life estimation. *ICASSP 2019–2019 IEEE International Conference on Acoustics* (pp. 3872–3876). Speech and Signal Processing (ICASSP): IEEE.
- Benker, M., Furtner, L., Semm, T., & Zaeh, M. F. (2021). Utilizing uncertainty information in remaining useful life estimation via Bayesian neural networks and Hamiltonian Monte Carlo. *Journal of Manufacturing Systems*, 61, 799–807.
- Caponetto, R., Rizzo, F., Russotti, L., et al. (2019). Deep learning algorithm for predictive maintenance of rotating machines through the analysis of the orbits shape of the rotor shaft. *International Conference on Smart Innovation* (pp. 245–250). Ergonomics and Applied Human Factors: Springer.
- Cardoso, V. H., de Oliveira Neto, G. C., da Silva, R. N. B., Alexandruk, M., Araujo, S. A. D., Tucci, H. N. P., Lourenço, S. R., Moraes, E. A. D., Vido, M., & Amorim, M. (2025). Enhancing production planning in metal-mechanical industries: statistical analysis and machine learning approach for predicting machine failures. *Journal of Industrial and Production Engineering*, 42(6), 571–586.
- Choo, Y. S., Lee, D. H., & Shin, S. J. (2025). Predicting remaining useful life and optimizing preventive maintenance for multistage manufacturing processes. *Quality and Reliability Engineering International*, 41(4), 1562–1576.
- Chung, W. H., Kremer, G. E. O., & Wysk, R. A. (2017). A dynamic programming method for product upgrade planning incorporating technology development and end-of-life decisions. *Journal of Industrial and Production Engineering*, 34(1), 30–41.
- Crespo, M. A., Candón, E., Gómez, J. F., & Serra, J. (2020). A comparison of machine learning techniques for LNG pumps fault prediction in regasification plants. *IFAC-PapersOnLine*, 53(3), 125–130.
- Dhini, A., Surjandari, I., Kusumoputro, B., & Kusiak, A. (2022). Extreme learning machine–radial basis function (ELM-RBF) networks for diagnosing faults in a steam turbine. *Journal of Industrial and Production Engineering*, 39(7), 572–580.

- Fan, Y., Nowaczyk, S., & Rögnvaldsson, T. (2020). Transfer learning for remaining useful life prediction based on consensus self-organizing models. *Reliability Engineering & System Safety*, 203, Article 107098.
- Fujimoto, S., Hoof, H., & Meger, D. (2018). Addressing function approximation error in actor-critic methods. *International conference on machine learning* (pp. 1587–1596) PMLR.
- Gu, G., Yao, Y., Pan, E., Chen, H., & Zhou, D. (2025). Multidimensional feature mining and selection for health indicator construction and remaining useful life prediction. *Quality and Reliability Engineering International*, 41(6), 2427–2440.
- Iyer, S. V., Sangwan, K. S., & Dhiraj, (2025). A cognitive digital twin for process chain anomaly detection and bottleneck analysis. *Journal of Industrial and Production Engineering*, 42(1), 65–87.
- Jeong, Y. E., Lee, Y. H., Kim, S. M., & Kim, Y. S. (2025). Remaining useful life prediction through the derivation of acceleration factors based on intermittent inspection data. *Quality and Reliability Engineering International*, 41(5), 1848–1860.
- Jiménez, H., Aribisala, A., Cavalcante, C., Do, P., & Lee, C. G. (2023). A deep reinforcement learning-based maintenance optimization for vacuum packaging machines considering product quality degradation. *Journal of Food Process Engineering*, 46(11), Article e14429.
- Kang, W., Jongbloed, G., Tian, Y., & Chen, P. (2024). Degradation index-based prediction for remaining useful life using multivariate sensor data. *Quality and Reliability Engineering International*, 40(7), 3709–3728.
- Kelley, J., & Hagan, M. T. (2024). Comparison of neural network narx and narmax models for multi-step prediction using simulated and experimental data. *Expert Systems with Applications*, 237, Article 121437.
- Kingma, D. P., & Welling, M. (2013). Auto-encoding variational bayes.
- Lee, J., & Mitici, M. (2023). Deep reinforcement learning for predictive aircraft maintenance using probabilistic remaining-useful-life prognostics. *Reliability Engineering & System Safety*, 230, Article 108908.
- Li, X., Ding, Q., & Sun, J. Q. (2018). Remaining useful life estimation in prognostics using deep convolution neural networks. *Reliability Engineering & System Safety*, 172, 1–11.
- Liu, Z. H., Meng, X. D., Wei, H. L., Chen, L., Lu, B. L., Wang, Z. H., & Chen, L. (2021). A regularized LSTM method for predicting remaining useful life of rolling bearings. *International Journal of Automation and Computing*, 18(4), 581–593.
- Ma, M., Sun, C., Mao, Z., & Chen, X. (2021). Ensemble deep learning with multi-objective optimization for prognosis of rotating machinery. *ISA transactions*, 113, 166–174.
- Men, C., Han, Y., & Huang, C. G. (2025). Prediction of equipment remaining useful life based on graph learning and spatiotemporal knowledge graph. *Quality and Reliability Engineering International*, 41(4), 1209–1224.
- Mitici, M., de Pater, I., Barros, A., & Zeng, Z. (2023). Dynamic predictive maintenance for multiple components using data-driven probabilistic RUL prognostics: The case of turbofan engines. *Reliability Engineering and System Safety*, 234, Article 109199.
- Mouhib, Z., Gallab, M., Lamrani Alaoui, Y., Merzouk, S., Di Nardo, M., El Bhiri, B., & Soulhi, A. (2025). Measuring total productive maintenance success: a new industrial evaluation using fuzzy SERVQUAL. *Journal of Industrial and Production Engineering*, 42(1), 14–29.
- Rasay, H., Azizi, F., Salmani, M., & Naderkhani, F. (2025). Joint optimization of condition-based maintenance and production rate using reinforcement learning algorithms. *Quality and Reliability Engineering International*, 41(3), 957–969.
- Ren, L. N., Li, K., Li, X., & Wang, Z. (2024). A novel equipment remaining useful life prediction approach considering dynamic maintenance threshold. *Quality and Reliability Engineering International*, 40(7), 4088–4103.
- Saxena, A., Goebel, K., Simon, D., et al (2008) Damage propagation modeling for aircraft engine run-to-failure simulation. In: 2008 international conference on prognostics and health management, IEEE, pp 1–9.
- Schulman, J., Wolski, F., & Dhariwal, P. (2017). Proximal policy optimization algorithms [arXiv:1707.06347](https://arxiv.org/abs/1707.06347) arXiv preprint.
- Su, C., Li, L., & Wen, Z. (2020). Remaining useful life prediction via a variational autoencoder and a time-window-based sequence neural network. *Quality and Reliability Engineering International*, 36(5), 1639–1656.
- Tanzeem, N., & Moghaddass, R. (2025). Deep variational Bayesian approach for sensor-driven modeling and monitoring of dynamic degrading systems. *Quality and Reliability Engineering International*, 41(6), 2346–2372.
- Ucar, M., & Qiu, R. G. (2005). E-maintenance in support of e-automated manufacturing systems. *Journal of the Chinese institute of industrial engineers*, 22(1), 1–10.
- Wang, L., Zhu, Z., & Zhao, X. (2024). Dynamic predictive maintenance strategy for system remaining useful life prediction via deep learning ensemble method. *Reliability Engineering & System Safety*, 245, Article 110012.

- Xanthopoulos, A. S., Chnitidis, G., & Koulouriotis, D. E. (2019). Reinforcement learning-based adaptive production control of pull manufacturing systems. *Journal of Industrial and Production Engineering*, 36(5), 313–323.
- Xiang, S., Qin, Y., Luo, J., Pu, H., & Tang, B. (2021). Multicellular LSTM-based deep learning model for aero-engine remaining useful life prediction. *Reliability Engineering and System Safety*, 216, Article 107927.
- Yan, Z. Z., Sun, L. J., & Liang, Y. Y. (2025). Remaining useful life analysis for two-stage nonlinear inverse Gaussian process with random effects. *Quality and Reliability Engineering International*, 41(4), 1447–1460.
- Yaqiong, L., Pan, Z., Yifan, L., & Xian, W. (2025). Dynamic predictive maintenance strategy for multi-component system based on LSTM and hierarchical clustering. *Quality and Reliability Engineering International*, 41(1), 30–50.
- Zhang, A., Qi, F., Fu, P., Yu, L., & Zhou, H. (2025). Condition-based maintenance policy for a multi-component system considering stochastic dependence and quality loss. *Quality and Reliability Engineering International*, 41(2), 625–640.
- Zhang, X. H., Zeng, J. C., & Gan, J. (2018). Joint optimization of condition-based maintenance and spare part inventory for two-component system. *Journal of Industrial and Production Engineering*, 35(6), 394–420.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Hima Soni¹ · Vibha Patel² · Arnav Modanwal³ · Hepil Italiya⁴ · Aastha Patel⁵

✉ Hima Soni
hima.k.soni@gmail.com

✉ Vibha Patel
vibhadp@vgecg.ac.in

¹ Research Scholar, Gujarat Technological University, Chandkheda, Ahmedabad, Gujarat 382424, India

² Information Technology Department, Vishwakarma Government Engineering College, Chandkheda, Ahmedabad, Gujarat 382424, India

³ Systems Engineer, Tata Consultancy Services, Garima Park, Gandhinagar, Gujarat 382010, India

⁴ AI/ML Engineer, MultiQoS, Gala Empire, Drive-In Road, Opp. Memnagar, Ahmedabad, Gujarat 380052, India

⁵ Business Analytics Department, MDI Gurgaon, Sukhrali, Gurgaon, Haryana 122007, India